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Antitumor and hepatoprotective effect of *Cuscuta reflexa Roxb.* in a murine model of colon cancer

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ABSTRACT

Ethnopharmacological relevance: Cuscuta reflexa Roxb. (*C. reflexa*) is a well-known traditional herbal plant, with numerous inherent therapeutic potentials including anticancer, antitumor, antibacterial, analgesic, anthelmintic, laxative and others. Moreover, the anticancer and antitumor potentials of this herb are ongoing with several trails, thus an attempt was made to assess the anticancer and hepatoprotective potentials of traditional *C. reflexa* herbs.

Method: The dried ethanolic extract of *C. reflexa* was tested for acute oral toxicity in the treated animals subsequently their behavioral, neurological, and autonomic profiles changes were observed. The preliminary anticancer effects of extracts against 1, 2- Dimethyl hydrazine (DMH) induced animals were assessed through barium enema X-ray, colonoscopy, and Aberrant crypt foci (ACF) studies. The blood samples of the animals (treated and untreated) were collected and their *in-vitro* histological parameters were evaluated by the experienced technician.

Results: It was observed that *C. reflexa* significantly reduced Disease activity indexing (DAI) level and ACF counting, as well as demonstrated similar activity as of the standard drug 5-Fluorouracil (5-FU). Histopathological results revealed that the apoptotic bodies decreased in the DMH-induced group (group II) during cancer progression while in 5-FU treated (group III) and *C. reflexa* treated (group IV and V) animals the apoptotic bodies were increased. Inversely, the mitotic bodies increased in group II animals and reduced in group III, IV, and V animals. In the colonic section, DMH-induced cancer assay exhibited significant effects on the levels of hemoglobin, Packed cell volume (PCV), Red blood cell (RBC) counts, Mean corpuscular hemoglobin concentration (MCHC), Mean corpuscular volume (MCV), and Mean cell hemoglobin (MCH), and was found to be less in group II animals whereas administration of *C. reflexa* efficiently recovered back the loss probably by healing the colon damage/depletion of cancer progression. Moreover, compared to the group II animals, the neutrophil count was within the normal range in *C. reflexa* administered group.

Conclusions: In the present study, the major hematological parameters significantly increased within DMH treated animals and exhibited extensive damage in the hepatic regions. Moreover, the histopathological findings demonstrated that the *C. reflexa* extracts potentially reduced the cell proliferation, with no toxicity. The *C. reflexa* extracts exhibited impending anti-cancer activity as well as protected the hepatic cells and thus could be potentially used in the management of colon or colorectal cancer and hepatic impairments.

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1. Introduction

Colorectal or colon cancer (CR/C cancer) is the malignancy of the intestinal epithelial cells that rapidly proliferate and mostly causes death if kept untreated. For this reason, CR/C cancer is considered one of the most alarming menaces worldwide (Hao et al., 2018). CR/C cancer is found to be the second most common cancer diagnosed in females and the third-most in males, furthermore it is the prime cause of cancer-mediated death (Ferlay et al., 2019; Das et al., 2020a). The diagnosis of CR/C cancers differs and depends upon the cancer phase, with an almost five-year existence rate (around 90% for phase-I) and around 10% for phase-IV patients. Surveys and case studies have shown that the patients treated with chemotherapy after surgeries, experienced cancer deterioration, metastatic illness and even death; emphasizing the insufficiency of the present state of therapeutic management and treatment approaches for this lethal malignancy (Weng and Goel, 2020; Barkat et al., 2020a). Earlier studies have demonstrated that the consequences of CR/C-mediated patients majorly depend upon the size of anatomical illness, including tumorous cells. It was also very distinct that none of the factor/s could significantly govern the diagnosis of this fatal disease alone (Grothey et al., 2018). In the last few years, there has been a drastic increase in the cognizance regarding the prevention of CR/C cancer and apart from conventional chemotherapy, various potential approaches (targeted drug delivery systems) have been developed for significant management and treatment of the patients affected by this disease (Das et al., 2020b). In addition to the toxic nature and excessive cost of the established modern remedies, the researchers have shown great interest in recognizing probable therapies based on natural products as they are safe and reasonably priced (Allen and Sears, 2019). Thus, natural product-based chemotherapy could be significantly used, either alone or with conventional chemotherapeutics for the effective management of CR/C cancer. Although for the effective management of CR/C cancers significant progress in targeted therapeutics has been perceived, still their clinical efficiency and clinical translations into marketed products are limited (Pearce et al., 2017; Barkat et al., 2020b). Moreover, dietary alterations, modernization of lifestyle habits, obesity, alcoholism, and smoking are some of the major causes of the surge in the frequency of CR/C cancers worldwide (Fitzmaurice et al., 2015).

Cuscuta reflexa Roxb. (*C. reflexa*), generally identified as Dodder (English), Amarabela (Hindi), is a natural herb that belongs to the Convolvulaceae family and has been reported with various medicinal properties. These species are usually found in the humid and tropical areas of India, Pakistan, Afghanistan, Malaysia, Nepal, and Thailand (Patel et al., 2012). The communities from countries like India, Nepal, and Bangladesh practice them to manage and cure tumors, pain, edema, and to maintain the hepatic system (Hossan et al., 2009; Siwakoti and Siwakoti, 2000). Even scientists have shown potential applications of herbal products modified with nanotechnological interventions (Barkat et al., 2021; Harshita et al., 2020). Additionally, this potential herb constitutes a variety of bioactive components such as quercetin, myricetin, kaempferol, linoleic acid, luteolin, astragalin, stearic acid, isorhamnetin, coumarin, β-sitosterol, oleic acid, palmitic acid, α-amyrin, sesamin, n-hentriacontane, and others (Vijikumar, 2011). Traditionally, *C. reflexa* has been reported to exhibit significant activities in the treatment of tumors, majorly in Indian and Chinese regions. In one of the studies, Chatterjee et al. evaluated the traditional antitumor efficacy of the ethanolic and chloroform extracts of *C. reflexa* against the Ehrlich ascites carcinoma (EAC) induced tumor in mice model. Results of the *in-vivo* and hematological parameters showed that the ethanolic and chloroform extracts of *C. reflexa* (200 mg/kg and 400 mg/kg) showed potential and significant antitumor activities against the treated animals. In this study, 5-fluorouracil was used as the standard antitumor drug. Administration of *C. reflexa* (200 mg/kg and 400 mg/kg) significantly reduced the total tumor volume and viable cell counts, but enhanced the non-viable cell counts and mean survival time, this enhanced the overall life span of the tumor-induced animal group. The different hematological parameters were also restored to normal levels. Therefore, all the results suggested that the chloroform and ethanolic extracts of *C. reflexa* exhibited significant anti-tumor potentials (Chatterjee et al., 2011).

Furthermore, various studies have been performed to investigate and demonstrate the anti-cancer activity of *C. reflexa* for the effective treatment of CR/C cancer or adenocarcinomas. Chatterjee et al., investigated the antitumor activity of the extracts (ethanolic and chloroform) of the entire *C. reflexa* plant that was orally administered (200 and 400 mg/kg) to the Swiss albino mice induced with EAC cells. Results showed that the extracts exhibited significant antitumor potential as they reduced the viable cell counts and tumor volume, rather enhanced the mean survival time and non-viable cell counts (Chatterjee et al., 2011). In another study, the extracts of *C. reflexa* exhibited anti-proliferative activities against human neuroblastoma (IMR-32) and human colon (502713) cancerous cell lines. Results of the comet assay revealed that cell death was probably due to apoptosis (ssDNA rupture). The extracts also exhibited antimicrobial activities and potentially affected both the gram-positive and gram-negative bacterial strains (Bhagat and Saxena, 2011). Riaz and other co-workers obtained four isolates from *C. reflexa* namely Scoparone (1), *p*-coumaric acid (2), stigmasta-3,5-diene (3) and 1-*O*-*p*-hydroxycinnamoylglucose (4). Results of the sulphorhodamine B assays revealed that amongst all these compounds, compound (4) exhibited potential anti-proliferative effects against HCT116 colorectal cells, while compounds (1) to (3) exhibited mild actions (Riaz et al., 2017). Although significant studies have been performed for evaluating the therapeutic potentials of *C. reflexa in-vitro* and *in-vivo*, however *in-vivo* studies for chemically induced animal models with histological details have not been reported yet. Thus, the present work was intended to determine various pharmacological activities exhibited by *C. reflexa* plant extracts including anticancer activity *in-vivo*, against chemically induced colorectal cancer in Wistar rats.

2. Materials and methods

2.1. Chemicals and reagents

1, 2- Dimethyl hydrazine (DMH) was purchased from Sigma-Aldrich Chemical Pvt Limited, Bangalore, India. 5-Fluorouracil (Fivoflu®), a commercial formulation (Fresenius Kabi, Dabur, India) was procured from the market. Serum glutamic-oxaloacetic transaminase (SGOT) diagnostic kit and Serum glutamic pyruvic transaminase (SGPT) estimation kit were purchased from ERBA, India. Sodium hydroxide (NaOH), sodium carbonate (Na_2CO_3), ethylene diamine tetraacetic acid (EDTA), Hydrogen Peroxide, Ethyl Alcohol, Glycerin, Benzidine, and Formaline (10%) were purchased from CDH Fine Chemicals, India. Barium sulfate suspension was purchased from Rankem, Gurgaon, India. All other chemicals were of analytical grade.

2.2. Procurement of plant and extraction

The whole plant was collected from the fields of Dhampur, Bijnor, Uttar Pradesh, India in June 2015. The plant specimen was then authorized by Dr. Anamika Tripathi from Hindu College Moradabad, India with reference no. HC/BOT/PERL-30 dated 22/07/15. The whole plant of *C. reflexa* was grounded to get the coarse powder and then subjected to successive extraction using chloroform with the help of a soxhlet extractor. After extraction with chloroform, the coarse powder of the *C. reflexa* plant was dried and further extracted with ethanol for 24 h. The liquid extract obtained was then cooled and placed on a water bath at temperature 40–50 °C until the entire solvent was evaporated. The dried extract obtained was weighed and the percentage yield was calculated using equation (1) concerning the crude drug. Further, the dried ethanolic extract was stored for pharmacological screening.

$$\text{Percentage yield} = \frac{\text{Weight of extracts obtained}}{\text{Weight of crude drug}} \times 100. \quad (1)$$

2.3. Experimental animals and pharmacological screening

The Institutional Animal Ethics Committee (IAEC) of College of Pharmacy, Teerthanker Mahaveer University, Moradabad (Reg. No. 1205/C/08/CPCSEA/21.04.08) approved animals were used for the current study. Healthy male Wistar albino rats (150–175 g) were utilized for performing in-vivo assays. Further, all the animals were housed individually and permitted for free contact with normal diets and water *ad-libitum*. The surgical interventions were carried out under sterile conditions. All the animals were acclimatized in standard laboratory circumstances (12 h light: 12 h dark) at room temperature (25±2 °C) for 10 days, preceding testing. The relative humidity was maintained between 50% and 60% and the body weight of each animal was checked weekly.

2.4. Acute oral toxicity

OECD-425 guidelines were followed to perform acute oral toxicity (OECD, 2008). The animals fasted overnight with water *ad-libitum*. The dosing was done in a stepwise process utilizing fixed dosages of 5, 50, 300, and 2000 mg/kg (exceptionally 5000 mg/kg). The preliminary dosing level was ascertained based on sighting studies. As the dose is estimated to gain some signs of toxicity deprived of causing extreme toxicity. Further, the animal groups were dosed at lower or higher doses, unless there was any sign of toxicity. This process was continued till the dosing caused any significant toxicity or demise of an animal was observed, or when higher dosing exhibits no significant effects or lower dosing causes the death of animals. The extract was given to animals in a single dose by gavage *via* a stomach tube or an appropriate intubation cannula. After dosing, each animal of different groups was observed exclusively at least once in the course of the first 30 min, intermittently during the first 24 h, with distinct considerations in the course of the first 4 h and days afterward, up to 14 days. Generally, the alterations in behavioral, neurological, and autonomic profiles were examined in the treated animals.

2.5. Selection of dose

After testing different doses of plant extract solution (5, 50, 300, and 2000 mg/kg body weight) the appropriate maximum tolerated safe dose was considered. 1/10th and 1/5th of the maximum tolerated safe dose was selected as the treatment dose for further studies of pharmacological activities according to OECD guidelines 425.

2.6. Anticancer assay (DMH-induced colorectal cancer study)

1 mM EDTA solution was formulated by dissolving 37.2 mg EDTA in 100 ml distilled water and 1 M NaOH by dissolving 4 g of sodium hydroxide in 100 ml. Accurately weighed quantity of DMH dissolved in 1 mM EDTA and the pH was adjusted to 6.5 with 1 mM NaOH to maintain the stability of the chemical.

Thirty-two male albino wistar rats were randomly categorized into two groups, one group of eight animals was treated with 1 mM EDTA and labeled as group I, the second group of twenty-four rats received a sub-cutaneous injection of DMH (20 mg/kg) once a week for consequently 4 weeks. After the DMH treatment is over, randomly two rats were selected and were sacrificed by decapitation method and their colon was subjected for the ascertainment of aberrant crypt foci (ACF) (Prasad, 1965). Further grouping of animals has been demonstrated in Table 1.

Table 1
Grouping of animals for pharmacological screening and anti-cancer assay.

Group	Group Type	Treatment	Total no. of animals	Dose
I	Positive Control	EDTA	08	1 mM
II	Negative Control	DMH + EDTA	06	20 mg/kg
III	Standard	5-FU + DMH	06	35 mg/kg
IV	Low dose	*C. reflexa* + DMH	06	200 mg/kg
V	High dose	*C. reflexa* + DMH	06	400 mg/kg

2.7. Determination of tumor induction (preliminary screening)

2.7.1. Barium enema X-ray

The barium sulfate solution enema is introduced into the sedated animal and at the same time, the animal abdominal X-ray is taken to detect the colon tumors. The faded part is presumed as colon tumors. Before starting the procedure, the animals were kept fasting for 24 h, and stomach wash was also done before 12 h of procedure (Tsunoda et al., 1992).

2.7.2. Colonoscopy

Under this dated condition an endoscope of 2–4 mm diameter was introduced to visualize the abnormal growth. Before the procedure the animals were kept fasting for 24 h, and stomach wash is also done before 12 h of procedure (Olson and Halbeg, 2011).

2.7.3. ACF determination (confirmatory study)

At the end of the fourth-week treatment with DMH, the colon of two selected animals was detached, washed with the help of 0.1 M potassium phosphate buffer saline (pH 7.2). The colon was opened longitudinally and placed over the filter paper by surfacing towards the luminal side. The dissected colon was dispersed in formalin (10%) for upto 12 h and cut into equal pieces of 2 cm long. Now every piece of the dissected colon segment was placed into the Petri dish having methylene blue (0.2–0.5%) for 2 min and excess dye was washed with phosphate buffer and examined under low magnification to count ACF. Scoring was done by resembling and counting the aberrant crypt from the normal crypt. Total counting of single, double, triple, and more than triple crypt in each segment was also taken into consideration for scoring. The aberrant crypt was traced by dark staining, thickened area and raised wall with extended slit-like lumen with enhanced distance from the lamina to the basal surface of cells (Sengupta et al., 2003).

2.8. Experimental design

After getting the confirmation report all DMH-treated animals were further categorized into four different groups (n = 6) (Table 1). Group I was given a standard pellet diet and served as the control, while group II to V animals was given a standard pellet diet with DMH (20 mg/kg), once a week, subcutaneously for the first four weeks. After four weeks, group III received 5-FU 35 mg/kg (A. A. Almagrami et al., 2014) while group IV and V animals received *C. reflexa* extract (200 mg/kg and 400 mg/kg respectively) along with DMH (20 mg/kg) once weekly for four more weeks. The treatment with *C. reflexa* extract and 5- FU was continued up to 16 weeks. At the end of the experiment, rats were kept on fasting overnight and then sacrificed by the decapitation method.

2.9. Disease activity indexing (DAI)

Disease activity indexing is a research tool that is usually applied to quantify the symptoms of the progression of the disease through scoring by which action of the drug can be presumed. With the help of benzidine, fecal occult blood test, body weight, and stool consistency were examined (Chen et al., 2007; Tian et al., 2013). DAI is the mutual values of weight loss, stool consistency, and bleeding. The values were categorized as:

i) For weight: 0 (no loss), 1(5–10%), 2 (10–15%), 3 (15–20%) and 4 (>20%) weight loss.
ii) For stool consistency: 0 (normal), 2(loose) and 4 (diarrhea).
iii) For bleeding: 0 (no blood), 2 (presence of blood) and 4 (gross blood).

2.10. Determination of ACF

At the end of 16 weeks, DMH and *C. reflexa* extract (200 mg/kg and 400 mg/kg) treated rat's colon was removed, washed, and flushed with the help of 0.1 M potassium phosphate buffer saline (pH 7.2) and for further followed the method described above in the section ACF determination (confirmatory) (Sengupta et al., 2003).

2.11. Estimation of hematological and histopathological parameters

Specimen collection and preparation: Blood samples were collected from the tail vein of the animal and were further used for the assessment of hemoglobin (Hb) level, RBC, and white blood cell (WBC) counts. All blood samples (3–4 ml) were collected in EDTA tubes. All the histological studies were performed by an experienced technician.

After imagining the crypt, the outline dyes were removed. The organized sections were arranged by hematoxylin and eosin (H & E) for sorting and a random selection was fixed in paraffin wax. Further, the paraffin blocks were consecutively cut into perpendicular sections (5 μm). The dissected section was stained with H&E and assessed for proliferative activities (Jucá et al., 2014).

2.12. Apoptotic bodies

After H & E staining, the slides were observed under the microscope. Also, twenty full longitudinal crypt sections of normal mucosa of rats were recorded for determining the existence of apoptotic cell resembled by cell contraction, loss of regular interaction with the adjacent cell, condensation of chromatin or creation of round/oval shaped nuclear cells (Caderni et al., 2000; Eggadi et al., 2014). The values of the apoptotic index can be calculated by using the formula depicted in equation (2).

$$Apoptoticbodies = \frac{\text{Number of apoptotic cells}}{\text{Cells scored}} \times 100 \tag{2}$$

2.13. Mitotic bodies

After H & E staining, the slides were visualized under the microscope. Also, twenty full longitudinal crypt sections of normal mucosa of the rat were recorded for determining the existence of mitotic bodies resembled by subsequent dark staining on H and E slides (hyperchromatism), dark-staining threads (Zheng et al., 1999).

2.14. Statistical analysis

The results of biochemical parameters were stated as mean value ± standard deviation (s.d). The variations within a set of data were evaluated by ANOVA studies, whereas the individual comparisons of group mean values were assessed by Dunnet's test (Sigma Stat 3.5). P values < 0.05 were considered to be statistically significant.

3. Results

3.1. Preparation of the extract and pharmacological screening

The extraction procedure was performed with the help of a soxhlet extractor. The percentage yield of coarsely powdered bark extracts of *C. reflexa* obtained by petroleum ether and ethanol were found to be 0.9% and 13.2% respectively. Further, the extracts were allowed for pharmacological screening.

3.2. Acute oral toxicity

The ethanolic extract of *C. reflexa* was subjected to acute oral toxicity assay as per the OECD-425 guidelines. The LD_{50} of the ethanolic extract of *C. reflexa* was calculated as per the OECD guidelines. Since no signs of toxicity or mortality was observed even at a dose of 2000 mg/kg, thus for the present study, 1/5th (400 mg) and 1/10th (200 mg) of maximum tested dose was selected as high dose and low dose samples respectively.

3.3. Confirmatory parameters for DMH induced colorectal cancer

There are various invasive methods available that confirm the chemically induced cancer in the large intestine but in this study, the confirmation of colorectal cancer induction was done by the use of non-invasive techniques such as Barium Enema X-ray and Colonoscopy. The result of these techniques confirms the induction of cancer in the colonic sections of the large intestine as shown in Fig. 1(A and B). The arrows (white) assigned in the figures represent the formation of tumors in the colonic region of the DMH-induced animals. The colonic sections showed abnormal areas in the smooth inner lining of the colon, indicating the formation of the cancerous microenvironment, which was further confirmed from the endoscopic studies performed using Colonoscopy.

3.4. Anti-cancer assays

In this assay, the effect of 5-FU, ethanolic extract of *C. reflexa* (low and high dose), and DMH were comparatively assessed for various blood components. The groups (treated animals) were considered as reported in Table 1. Also, the details of these studies have been summarized in Fig. 2 and detailed below.

3.5. Hematological results

3.5.1. Effect on RBC count

The comparative effect of different treatments over the RBC count is represented in Fig. 2(a). The RBC count was measured as millions per cubic millimeter (million/mm^3). In contrast to the group I animals, the RBC count was reduced by 30% in animals belonging to group II while animals from groups III, IV, and V exhibited an improved RBC count by 21.98%, 25.98%, and 18.75% respectively.

3.5.2. Effect on mean corpuscular hemoglobin concentration (MCHC)

MCHC is the mean hemoglobin concentration present in a specified volume of RBCs, or in other words, the ratio of hemoglobin mass to the volume of RBCs. The MCHC is measured as grams/deciliter (g/dl).The comparative effect of different treatments over the MCHC is represented in Fig. 2(b). In contrast to the group I animals, group II animals have a decreased level of MCHC by 1%. Meanwhile, as compared to group II, the MCHC was increased by almost 2% in both group III and IV animals. Also, group IV and V animals recovered back the MCHC by 1.6% and 0.6% respectively.

3.5.3. Effect on mean cell hemoglobin (MCH)

The MCH value is the average mass of hemoglobin present per RBC of a blood sample. The MCH is measured as picograms/cell (pg/cell). The

Fig. 1A. Detection of tumors formation by Barium Enema X-ray (arrows signifying tissue aberration and tumor formation).

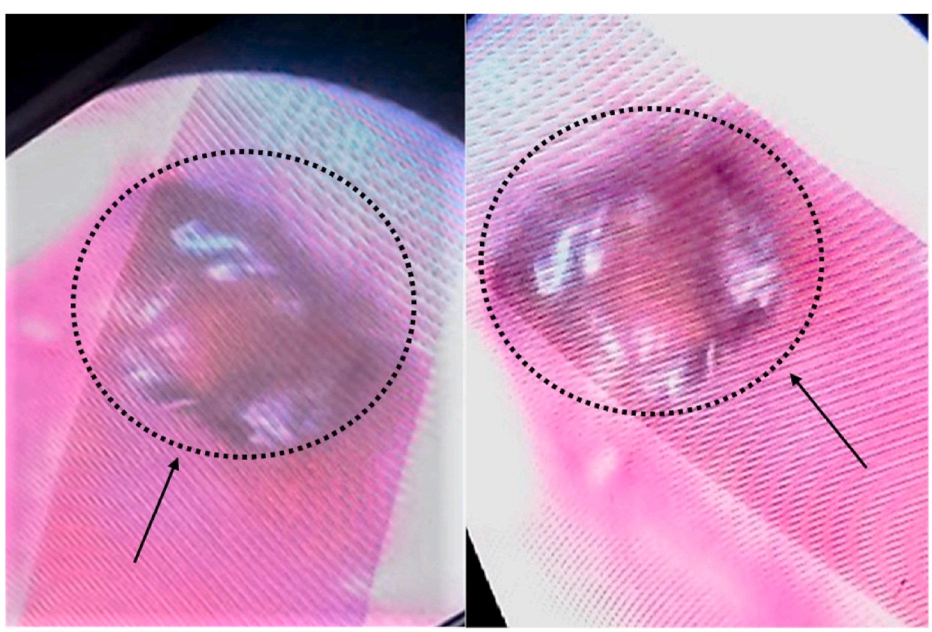

Fig. 1B. Detection of tumors formation with colonoscopy (arrows signifying disrupted lining of the smooth surface of colonic sections).

comparative effect of different treatments over the MCH value is represented in Fig. 2(c). In this study, group II animals showed a reduction in the MCH value by 9.27%, which was insignificant in comparison to the group I animals. Meanwhile, compared to the group II animals, group III, IV, and V animals exhibited increased values of MCH with 3.4%, 1.88% and 0.67% respectively.

3.5.4. Effect on mean corpuscular volume (MCV)

The MCV is the mean concentration of RBCs, could be directly calculated by an automated hematology analyzer, or could be measured from hematocrit (Hct) and the RBS counts. The comparative effect of different treatments over the MCV value is represented in Fig. 2(d) and is measured as femtoliters/cell (fl/cell). Furthermore, in comparison to the group I animals, group II animals showed a reduction in the MCV by 8.08%. In addition, the group III, IV, and V animals exhibited no

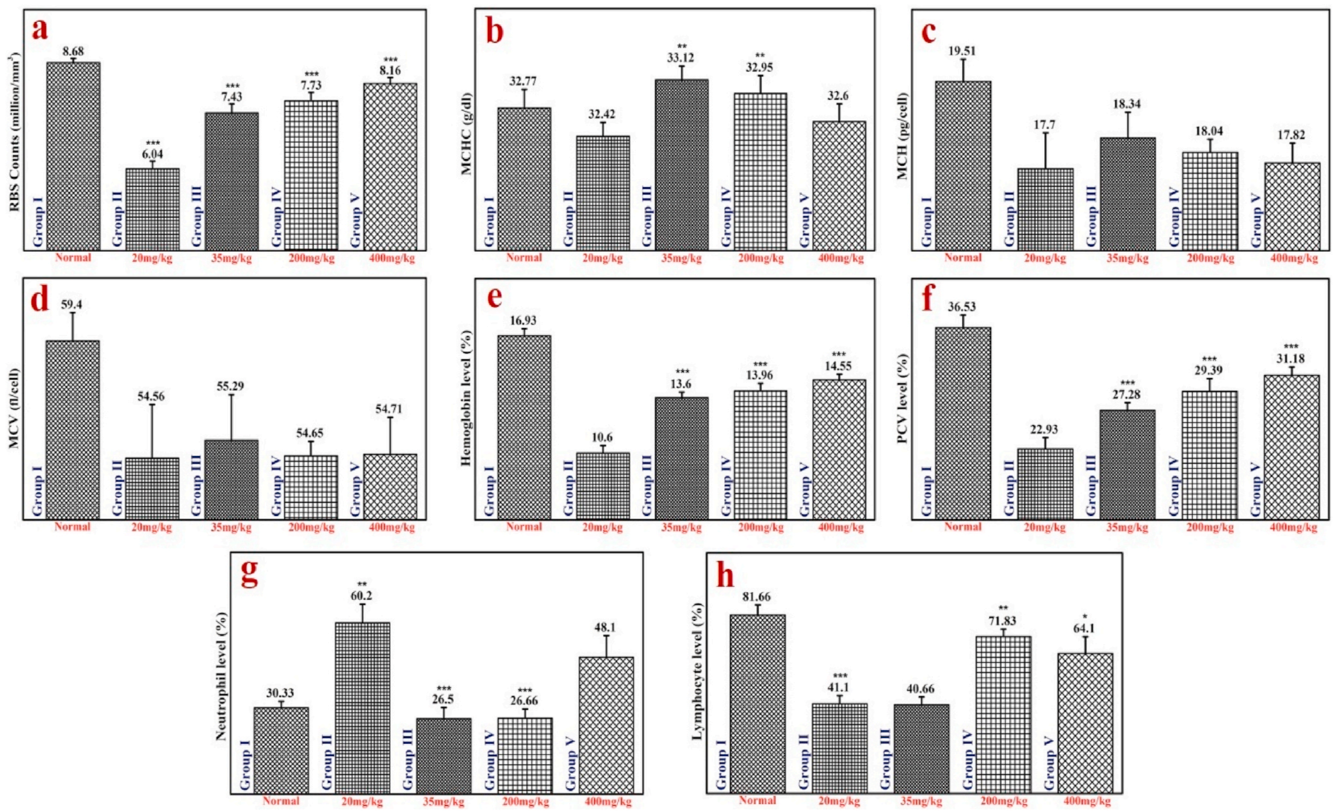

Fig. 2. Effect of 5-FU, ethanolic extract of *C. reflexa* and DMH on various blood components-a) Effect on RBC Count; b) Effect on mean corpuscular hemoglobin concentration (MCHC); c) Effect on mean cell hemoglobin (MCH); d) Effect on mean corpuscular volume (MCV); e) Effect on Hemoglobin level; f) Effect on packed cell volume (PCV); g) Effect on Neutrophil level; h) Effect on Lymphocyte. Values are represented as mean ± s.d. (n = 6) one-way ANOVA. Where, * represents significant at $p < 0.05$, ** represents highly significant at $p < 0.01$, and *** represents very significant at $p < 0.001$. All values were compared with negative control.

significant effects, and also the resultant level of MCV increased by 1.32%, 0.16%, and 0.002% respectively (insignificant increase), showing negligible effects over the size and volume of the RBC.

3.5.5. Effect on Hemoglobin level

The comparative effect of different treatments over the hemoglobin level is represented in Fig. 2(e) and is measured in percent (%). It was observed that the induction of colon malignancy by DMH caused a reduction in Hemoglobin levels. In comparison to the group I animals, group II animals exhibited a 37.2% reduction, while group III, IV, and V animals showed increased hemoglobin levels by 22.7%, 24.14%, and 27.14% respectively. It was concluded that all the treated groups except group II animals exhibited an increase in hemoglobin levels.

3.5.6. Effect on packed cell volume (PCV)

The comparative effect of all the samples over PCV values is represented in Fig. 2(f) and is measured in percent (%). In comparison to the group I animals, the PCV values reduced by 37.22% in group II animals, while group III, IV, and V animals manifested a significant increase in the PCV values by 18.70%, 21.98%, and 26.45% respectively. Results showed that all the treated groups except group II animals had an increased level of PCV.

3.5.7. Effect on neutrophil level

The comparative effect of all the samples over the neutrophil level is expressed in Fig. 2(g) and the results are shown in percent (%). The obtained results showed that the level of neutrophil in animals from groups III, IV, and V, significantly improved and returned to the normal range. Moreover, in group III and IV animals, a significant reduction (55.9% and 55.72%) was observed in the level of neutrophil (increased due to DMH administration) while group V animals showed no significant reduction in this case. Thus, as compared to the lower dose of group *C. reflexa* treated animals, higher dose-treated animals showed better results, indicating that the treatment followed a dose-dependent behavior.

3.5.8. Effect on lymphocyte

The comparative effect of all the samples over the lymphocyte level is represented in Fig. 2(h) and the results are expressed in percent (%). Group III and IV animals showed no significant increment in lymphocytes as compared to the group I treated animals, while deduction (49.79%) was observed in group II animals. Thus, as compared to group II and III results, the *C. reflexa* treated animals showed better results, as they maintained the lymphocyte levels.

3.5.9. Effect on aberrant crypt foci (ACF)

The images of the formation of ACF in the colonic environment and the comparative effect of all the samples over ACF are summarized in Fig. 3(A) and (B) respectively. ACF was identified stereoscopically discrete from the normal crypt by darker staining, greater size, oval shape, denser epithelial lining, and greater pericryptal zone. Administration of 5-FU significantly reduced the single, double and triple crypts and more than three crypts as compared to the DMH-induced group (group II animals). In comparison to the group II animals, group III animals exhibited significant inhibition in single ACF by 44.05%, double ACF by 44.97%, triple ACF by 43.49% and more than three ACF by 54.7%. Moreover, group IV, and V animals showed a significant reduction in all ACF with similar behavior as standard group (untreated) animals, represented in Fig. 3(B).

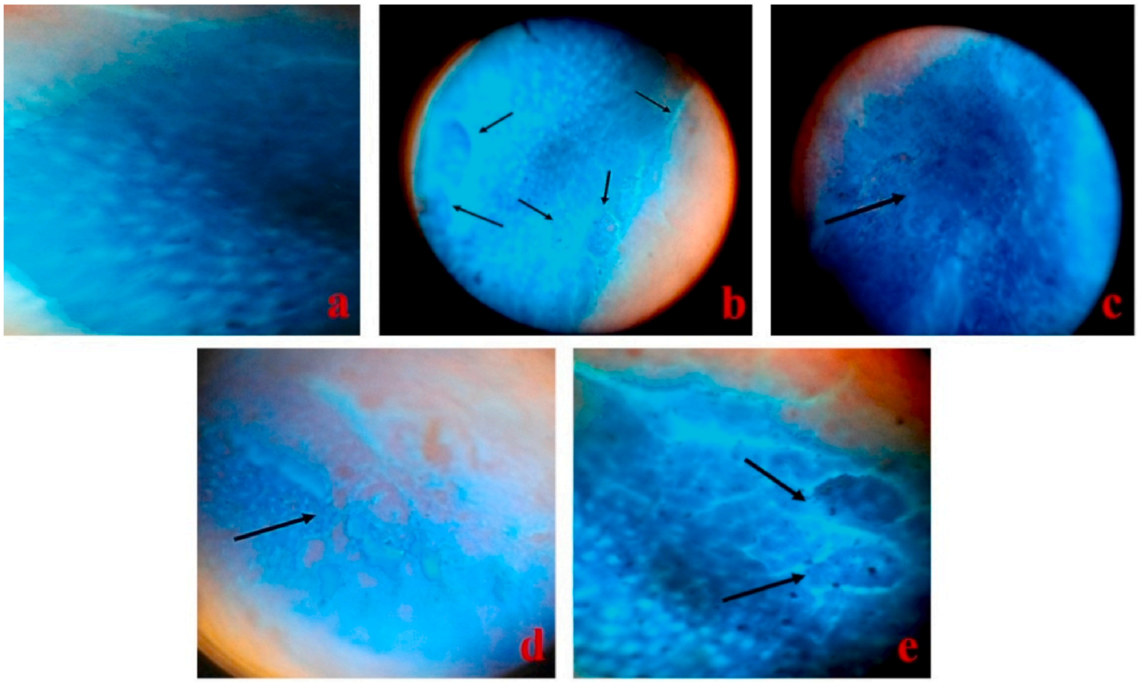

Fig. 3A. The images of the formation of ACF (indicated with arrows) in the colonic environment and the comparative effect of all the samples over ACF- a) Normal colon crypt; b) ACF in DMH induced Group; c) ACF in DMH +5 FU treated group; (d) ACF in DMH + *C. reflexa* extract 400 mg/kg treated group; (e) ACF in DMH + *C. reflexa* extract 200 MG/kG treated group.

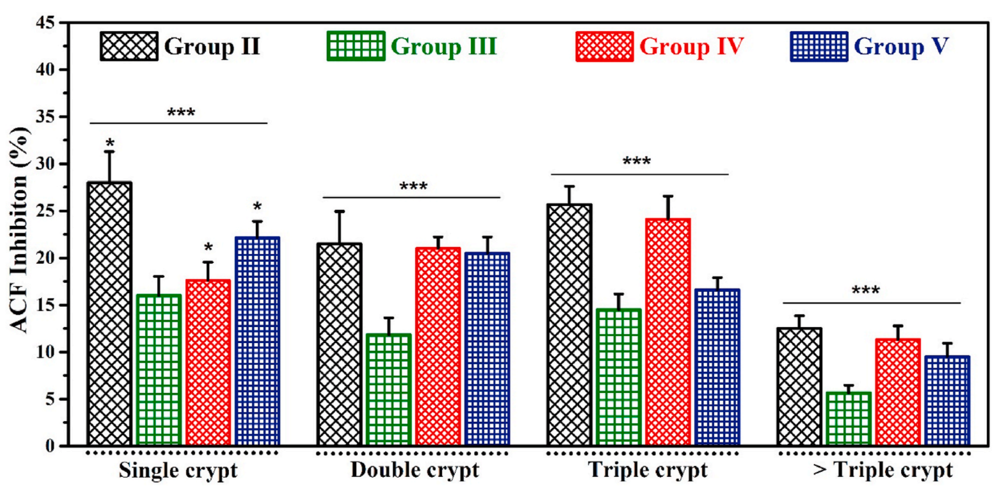

Fig. 3B. Effect of 5-FU, ethanolic extract of *C. reflexa* and DMH on aberrant crypt foci (single, double, triple and more than triple crypts). Values are represented as mean ± s.d (n = 6) one-way ANOVA. Where, * represents significant at $p < 0.05$, and *** represents very significant at $p < 0.001$. All values are compared with negative control.

3.6. Effect on apoptotic activity index (AAI)

The effect of carcinogenic DMH and ethanolic extract of *C. reflexa* along with standard drug 5-FU is represented in Fig. 4(a). Group II animals exhibited reductions in apoptotic indexing by 23.37% compared to the animals of group I. Furthermore, group III animals showed the tendency to increase the number of apoptotic cells and manifested a significant enhancement in apoptotic indexing of 51.68% in contrast to group II animals. Moreover, group IV and V animals exhibited a surge in apoptotic indexing by 28.79% and 17.85%, respectively. Although, the results obtained for group V animals were not significant when compared with group II animals. It has been observed that group V exhibited a lower apoptotic index as compared to group IV, showing that the apoptosis reduction demonstrated dose-dependent behavior. Also, the group IV results were comparatively similar to that of group II (positive control) values.

3.7. Effect on mitotic activity index (MAI)

The effect of carcinogenic DMH and ethanolic extract of *C. reflexa* along with standard drug 5-FU is illustrated in Fig. 4(b). In group II animals, an increase of 24.31% in mitotic indexing was observed when compared with group I animals. As compared to group II animals, the mitotic indexing was significantly reduced by 20.45% and 18% in group III and V animals, respectively.

Furthermore, group IV animals showed no significant reduction in mitotic indexing. Also, the histopathological finding of apoptotic bodies and mitotic bodies is sown in Fig. 5.

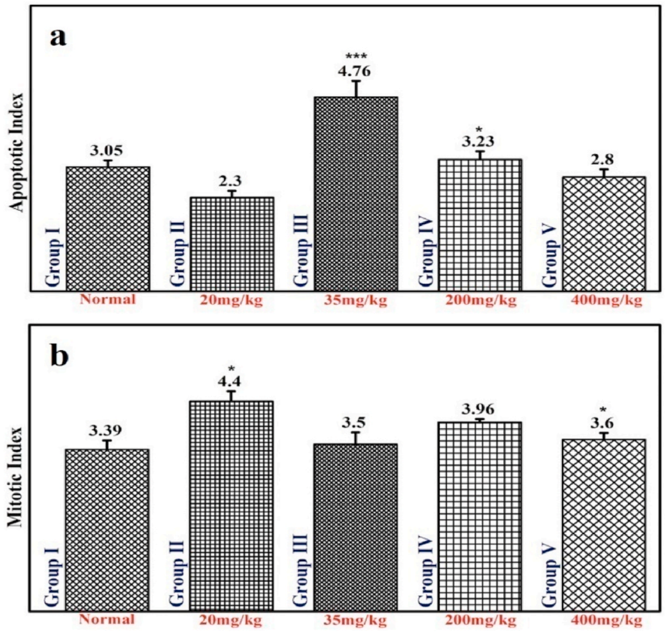

Fig. 4. a) Effect of 5-FU, Ethanolic extract of *C. reflexa* and DMH on Apoptotic Activity Index (AAI); b) Effect of 5-FU, Ethanolic extract of *C. reflexa* and DMH on Mitotic Activity Index (MAI). Values are represented as mean ± s.d (n = 6) one-way ANOVA. Where, * represents significant at $p < 0.05$, and *** represents very significant at $p < 0.001$. All values are compared with negative control.

3.8. Effect on SGPT and SGOT levels

The effect of carcinogenic DMH and ethanolic extract of *C. reflexa* along with standard drug 5-FU over DAI is illustrated in Fig. 6. Results showed that the treatment with 5-FU, 200 mg/kg, 400 mg/kg decreased the SGPT level, compared to the negative control group. On the other hand, SGOT level significantly increased when treated with the DMH group, however administration of *C. reflexa* extract (200 mg/kg and 400 mg/kg) exhibited no significant reduction in the increased SGOT level.

3.9. Disease activity index (DAI)

The effect of carcinogenic DMH and ethanolic extract of *C. reflexa* along with standard drug 5-FU over DAI is represented in Fig. 7. Generally, the DAI is performed by weight variation, stool consistency, and fecal occult blood test. The evaluation is performed on the progression of the disease condition. The DAI level of group II animals was compared with all other groups, in which all the groups showed a significant reduction in DAI. Among them, group III and V animals decreased DAI equally and significantly by 63.23% and group IV animals decreased DAI by 54.41%.

3.10. Histopathological study

The results of the histopathological examination are demonstrated in Fig. 8. The normal colonic section showed no significant proliferation-related parameters like typical ACF, hyperplastic ACF, adenomatous ACF, or some dysplastic foci (Fig. 8(a and b)). Furthermore, the colonic section of DMH-treated rats (Fig. 8(c and d)) showed typical ACF, hyperplastic ACF, adenomatous ACF, and some dysplastic foci. Whereas, as compared to the DMH induced group, less proliferation including typical ACF, hyperplastic ACF was found less in the group treated with standard drug 5-FU (Fig. 8(e and f)). Also, as compared to the DMH-induced group, low (Fig. 8(g)) and high (Fig. 8(h)) doses of *C. reflexa* significantly reduced the cell proliferation.

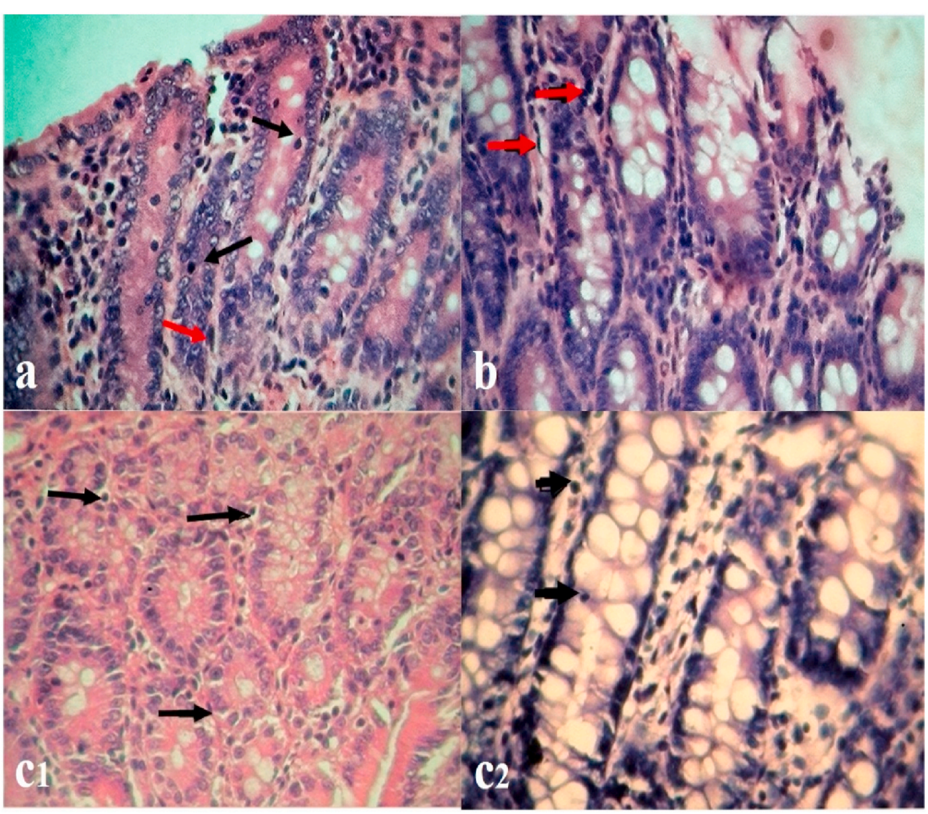

Fig. 5. Histopathology finding of apoptotic bodies and mitotic bodies – a) Red arrow showing apoptotic bodies, black arrow showing mitotic bodies in 5-FU group; b) Arrow indicating mitotic bodies in DMH induced group; C1) Arrows indicating apoptotic bodies in *C. reflexa* (200 mg/kg) treated group; C2) Arrows indicating apoptotic bodies in *C. reflexa* (400 mg/kg) treated group. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

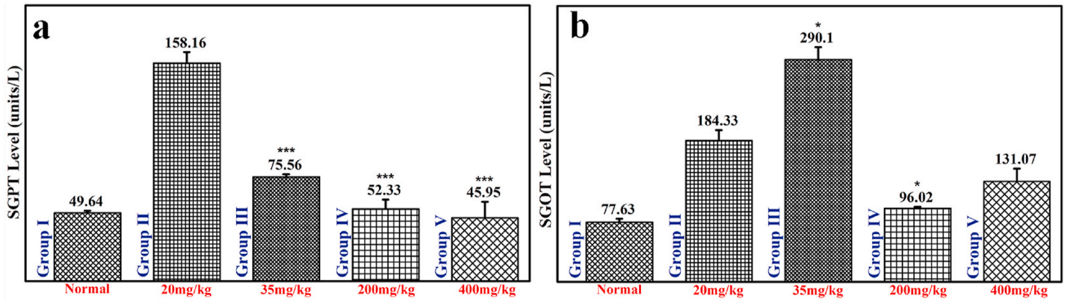

Fig. 6. Effect of 5-FU, DMH, ethanolic extract of *C. reflexa* (200 mg/kg and 400 mg/kg) on SGPT and SGOT levels. Values are represented as mean ± s.d (n = 6) one-way ANOVA. Where, * represents significant at $p < 0.05$, and *** represents very significant at $p < 0.001$. All values are compared with negative control.

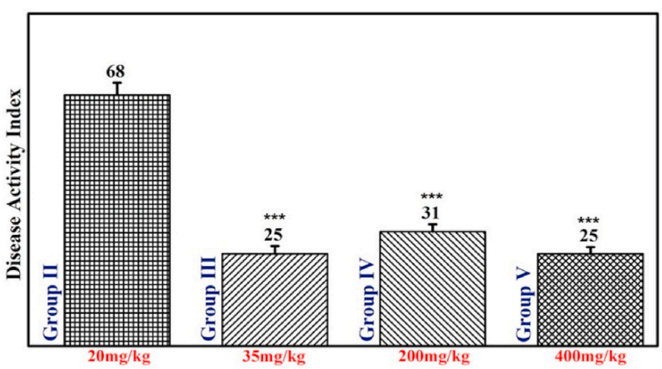

Fig. 7. Effect of 5-FU, DMH, ethanolic extract of *C. reflexa* (200 mg/kg and 400 mg/kg) on Disease Activity Index (DAI).
Values are represented as mean ± s.d (n = 6) one way ANOVA. Where, *** represents very significant at $p < 0.001$. All values are compared with negative control.

4. Discussions

Colorectal or colon cancer (CR/C cancer) is related to the malignancies in the epithelial cells of the intestinal regions and is one of the most commonly diagnosed diseases in both females and males worldwide (Hao et al., 2018). The 1, 2- dimethyl hydrazine (DMH) has been reported for its carcinogenic activities specifically in the colon region, as it triggers the production of oxidative stress and methyl free radicals. The oxidative stress further advances into chronic inflammatory responses, initiating CR/C cancer (Terzić et al., 2010). Additionally, research on the chemotherapeutic potentials of natural or herbal products has gained immense considerations due to their pharmacological significance in resolving different human ailments related to oxidative stress and inflammations. *C. reflexa* is generally identified as a natural herb that belongs to the Convolvulaceae family and has been reported with various therapeutic activities, including anticancer, antibacterial, antioxidants, antiviral, antispasmodic, antihypertensive, anticonvulsant, and hemodynamic (Tanruean et al., 2019). Furthermore, as per the earlier literature, the anti-CR/C activity and hepatoprotective activity of *C. reflexa* ethanolic extracts against DMH-induced Wistar rats have not been reported yet and for the first time it is reported in this experimental work.

In our study, the ethanolic extract of *C. reflexa* was obtained and was further subjected to acute oral toxicity assay in the Wistar rats (animal model) as per the OECD-425 guidelines. The oral toxicity assay results showed that the treated animals exhibited no toxicity or mortality even at a higher dose (2000 mg/kg). Further, in the selected animals, cancer was induced with DMH and the induction of cancer in the intestinal regions was confirmed through Barium Enema X-ray and Colonoscopy. The potential anti-CR/C and hepatoprotective activity of *C. reflexa* ethanolic extracts (low and high dose) against the DMH-induced Wistar

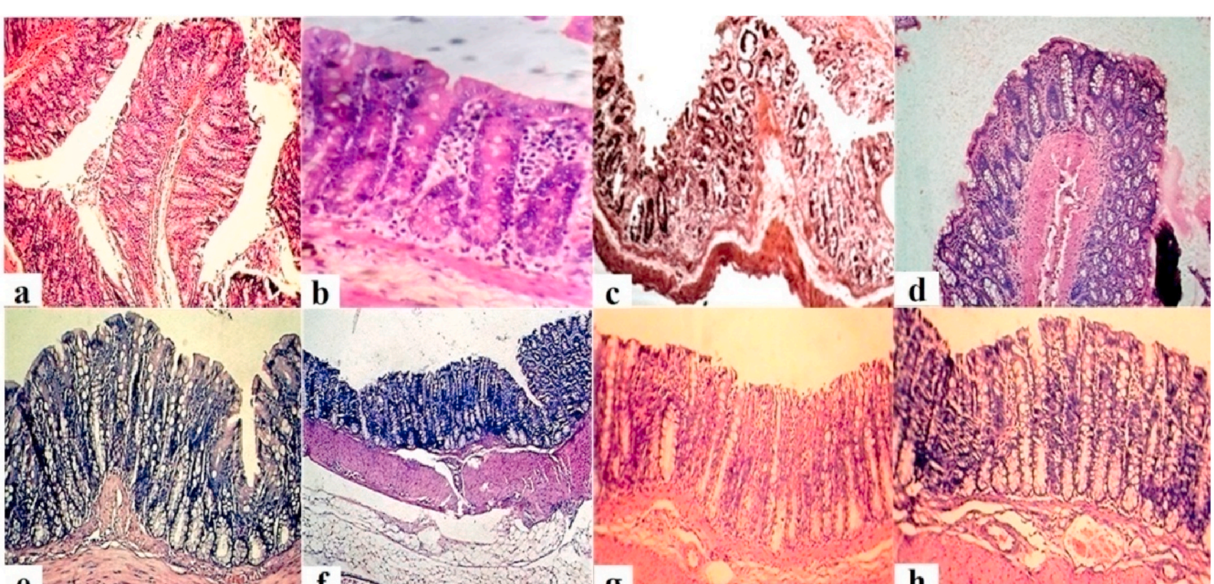

Fig. 8. The histopathological results and images: H & E staining of normal colon (a & b); H & E section of DMH-induced group (c & d); H & E staining DMH + 5-FU (35 mg/kg) treated (e & f); H & E section of DMH + *C. reflexa* (200 mg/kg) (g) and *C. reflexa* (400 mg/kg) (h) treated groups.

rats were assessed through histological and histopathological parameters.

Many studies have been reported earlier which showed that treatment of animals with DMH demonstrated severe alterations in various blood parameters such as level of RBC counts, MCHC, MCH, MCV, hemoglobin, PCV, neutrophil, lymphocyte. In one of the studies by Kudatarkar and Nayak (2018), the potential effect of *Cassia alata* extract (CAE) was studied against DMH-induced female rats. The results of their hematological parameters showed that CAE-treated animals exhibited a decrease in WBC level as compared to DMH-induced animals, which meant that CAE significantly neutralized WBC level. Also, as compared to the DMH-induced animals, the CAE-treated animals showed a severe decrease in RBC level. In this study, the hematological investigation was performed and compared amongst all the animal groups. The RBC count was reduced in DMH-induced animals which were found to be improved in animals treated with 5-FU and *C. reflexa* extracts (lower and higher doses). The MCHC, MCH and PCV levels were found to be decreased in DMH-induced animals, however it was increased in the 5-FU and *C. reflexa* extracts (lower and higher doses) treated animals. Also, *C. reflexa* extracts (lower and higher doses) treated animals significantly recovered the values of MCHC. The MCV level was found to be decreased in DMH-induced animals while 5-FU and *C. reflexa* extracts (lower and higher doses) treated animals showed no significant effects. The hemoglobin level was significantly reduced due to the induction of colon carcinogenesis by DMH. The hemoglobin level was found to be increased in all the animal groups except the DMH-induced animal group. The DMH-induced animals showed a drastic decrease in the neutrophil and lymphocyte levels. Moreover, the animals treated with 5-FU and *C. reflexa* extracts (lower and higher doses) significantly improved the levels of neutrophil and lymphocyte, and returned to normal conditions. The carcinogenic effects of DMH over hematological parameters were in accordance with an earlier reported study by Byelinska et al. (2015). In this study, the researchers investigated the morphological and functional conditions of blood cells in the DMH-induced colon cancer in rat models. Results showed that the DMH-induced animals exhibited reduced levels of MCH and MCHC, while increased the reticulocytes number and indirect bilirubin. These findings indicated hemolysis of RBCs because of the treatment of DMH during the development of tumors in rat models. Also, the level of monocytes, platelets, eosinophilic granulocytes was increased considerably.

Mori et al. (2005) reported that after the administration of DMH (carcinogen), the aberrant crypt foci (ACF) start appearing in a moderately shorter time and they persisted for a longer duration in the bowel mucosa. Earlier studies have demonstrated that the ACF was used as a biomarker to assess the initiation and elevation of chemically-induced carcinogenesis within *in-vivo* models (Fontana et al., 2001; Perše and Cerar, 2011). In this study, the ACF was witnessed in all the experimental animal groups (Fig. 3A), more specifically in the control group. The DMH-treated group exhibited a greater prevalence of ACF with bigger crypts as compared to the *C. reflexa* ethanolic extract-treated group (Fig. 3A and B). Orlando et al. (2008) mentioned that the probability of development of colon cancer increases significantly with an increase in the number of ACF and crypt multiplicity. The animals treated with *C. reflexa* ethanolic extract exhibited lesser crypt multiplicity as compared to the animals treated with DMH. Similar results were observed in the work of A.A. Almagrami et al. (2014), where it was reported that the oral administration of 250 and 500 mg/kg/day *Acanthus ilicifolius* significantly decreased the ACF count in the colon of azoxymethane-induced rats. Moreover, in our study we have demonstrated that the 5-FU and *C. reflexa* (lower and higher dose) treated animals showed similar behavior and significantly reduced all types of ACF (single, double, triple and more than three crypts) as compared to DMH-induced animals (group II animals).

Numerous reported studies showed that DMH-induced colon cancer in the rat model exhibited crypt proliferation and dysplasia in the animal colonic tissues. The histopathological studies of the sections of colonic tissues in the control group, and the *C. reflexa* extracts (lower and higher dose)-treated animals demonstrated normal histological structures of the mucosa, however, the DMH-treated animals' demonstrated diverse kinds of dysplasia (Figs. 5 and 7). These results were according to previous findings as reported by Ansil et al. (2013), Reynoso-Camacho et al. (2015) and López-Mejia et al. (2019). As per our investigations, the histopathological studies demonstrated that the supplementation of *C. reflexa* extracts (lower and higher dose) significantly affected the phases induced by DMH in the colonic sections of the animals. The values of AAI and MAI were found to be enhanced in the DMH-treated animals and the values were potentially reduced with the treatment of *C. reflexa* extracts as well as the standard drug (5-FU). The colonic sections of DMH-treated rats demonstrated typical ACF, hyperplastic ACF, adenomatous ACF, and some dysplastic foci while, *C. reflexa* (higher and lower dose) treated animals showed negligible counts of typical ACF and hyperplastic ACF. As compared to the DMH-induced group, the *C. reflexa* (higher and lower dose) treated animals significantly reduced cell proliferation. Also, the colonic sections showed that the *C. reflexa* extracts (lower and higher dose) extensively altered the efficiency at which DMH instigated the histological alterations. The DMH-treated sections of colonic tissues exhibited well-distinguished signs of dysplasia and further treatment of these tissues with *C. reflexa* extracts (lower and higher dose) significantly repaired the typical histological constructions in the colonic tissues, devoid of any evident signs of dysplasia. Also, from the results of the SGOT and SGPT studies, it was confirmed that the *C. reflexa* (higher and lower dose) treated animals significantly reduced the elevated levels of both SGOT and SGPT. Overall, the *C. reflexa* extracts exhibited hepatoprotective behavior. Similar results were observed in a study where the researchers showed the anti-carcinogenesis activities of selenium metal over the DMH-induced animals (Ghadi et al., 2009). Also, the capability of *C. reflexa* extracts (lower and higher dose) to reinstate the histological modifications indicated the anti-CR/C potentials of these herbal extracts.

Conclusively, the current study showed that the ethanolic extracts of the *C. reflexa* exhibited significant pharmacological activity over the chemically induced colon cancer in Wistar rats. The animals were treated with DMH and further, noninvasive techniques such as Barium Enema X-ray and Colonoscopy assisted to confirm the formation of abnormal growth in the colonic section. The hepato-specific and major blood biochemical parameters such as RBC count, level of hemoglobin, PCV, MCHC, MCV, MCH in animal models were significantly affected when treated with DMH (group II animals), 5-FU (group III animals), and ethanolic extracts of *C. reflexa* (group IV and V animals). In detail, group IV and V animals showed decreased levels of the DAI and ACF counting, which was found to be similar to the results of group III animals. The AAI decreased in the animals belonging to group II and was significantly increased in animals that belong to groups III, IV, and V, meanwhile, the MAI increased in animals that belong to group II and decreased in animals that belong to groups III, IV and V. Thus, it was observed that group IV and V animals significantly increased the apoptotic bodies counting and decreased the mitotic bodies count. DMH-induced animals showed intestinal bleeding which was due to cancer progression or damage or tumor formation and was confirmed by benzidine fecal occult stool test. After administering *C. reflexa* and 5-FU, the benzidine test of the same group became negative, thus it was proved that *C. reflexa* exhibited significant protective activity against DMH-induced colon damage in Wistar rats. Furthermore, due to intestinal bleeding, the blood comes into the stool and the level of hepato-specific/major blood biochemical parameters (mentioned above) were found to be less in the DMH-induced group whereas administration of *C. reflexa* recovered the loss, maybe due to healing the colonic damage or depletion of cancer progression. During cancer progression, high levels of neutrophils were reported. Importantly, the neutrophil to lymphocyte ratio was presented as a prescient factor in various types of tumors, including colorectal cancers. In group II animals, the neutrophil count was found to be higher than the normal range, whereas, in group IV and

V animals, it was found to be in the normal range. Group II animals demonstrated significant hepatic damage, witnessed from the increased levels of hepato-specific and major blood biochemical parameters. However, *C. reflexa* administration by DMH-induced group animals showed reduced levels of SGPT and SGOT. Conclusively, the anticancer and histopathological studies revealed that *C. reflexa* exhibited a potential and protective role against DMH-induced colon cancer and hepatic damage.

CRediT authorship contribution statement

Shobhit Mishra: Writing – original draft, design and executed the experimental work. All authors contributed to the paper. All authors have seen and approved the revised version of the manuscript being submitted. They warrant that the article is the authors' original work, hasn't received prior publication and isn't under consideration for publication elsewhere. **Fahad Saad Alhodieb:** Formal analysis, contributed in the statistical analysis and revision of the manuscript. All authors contributed to the paper. All authors have seen and approved the revised version of the manuscript being submitted. They warrant that the article is the authors' original work, hasn't received prior publication and isn't under consideration for publication elsewhere. **Md. Abul Barkat:** Formal analysis, Writing – original draft, design and executed the experimental work. analyzed the data and wrote the manuscript. All authors contributed to the paper. All authors have seen and approved the revised version of the manuscript being submitted. They warrant that the article is the authors' original work, hasn't received prior publication and isn't under consideration for publication elsewhere. **Mohd. Zaheen Hassan:** Formal analysis, Writing – original draft, design and executed the experimental work. analyzed the data and wrote the manuscript. All authors contributed to the paper. All authors have seen and approved the revised version of the manuscript being submitted. They warrant that the article is the authors' original work, hasn't received prior publication and isn't under consideration for publication elsewhere. **Harshita Abul Barkat:** Formal analysis, Writing – original draft, analyzed the data and wrote the manuscript. All authors contributed to the paper. All authors have seen and approved the revised version of the manuscript being submitted. They warrant that the article is the authors' original work, hasn't received prior publication and isn't under consideration for publication elsewhere. **Raisuddin Ali:** Formal analysis, Writing – original draft, analyzed the data and wrote the manuscript. All authors contributed to the paper. All authors have seen and approved the revised version of the manuscript being submitted. They warrant that the article is the authors' original work, hasn't received prior publication and isn't under consideration for publication elsewhere. **Perwaiz Alam:** All authors contributed to the paper. All authors have seen and approved the revised version of the manuscript being submitted. They warrant that the article is the authors' original work, hasn't received prior publication and isn't under consideration for publication elsewhere. **Ozair Alam:** Formal analysis, Writing – original draft, analyzed the data and wrote the manuscript. All authors contributed to the paper. All authors have seen and approved the revised version of the manuscript being submitted. They warrant that the article is the authors' original work, hasn't received prior publication and isn't under consideration for publication elsewhere.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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List of abbreviations

5-FU 5-Fluorouracil
C. reflexa Cuscuta reflexa Roxb
AAI Apoptotic activity index
ACF Aberrant crypt foci
Bcl-2 B-cell lymphoma 2
COX-2 Cyclooxygenase-2
CR/C cancer Colorectal cancer
DAI Disease activity indexing
DAPI 4′,6-Diamidino-2-Phenylindole (double stranded DNA staining)
DMH 1, 2- Dimethyl hydrazine
EAC Ehrlich ascites carcinoma
EDTA Ethylenediamine tetraacetic acid
MAI Mitotic activity index
MCH Mean cell haemoglobin
MCHC Mean corpuscular hemoglobin concentration
MCV Mean corpuscular volume
MTT 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide
NaOH Sodium hydroxide
Na_2CO_3 OECD guidelines 425 Sodium Carbonate Organisation for Economic Co-operation and Development guidelines 425
PCV Packed cell volume
RAW264 Murine macrophage cells
RBC Red blood cell
SGOT Serum glutamic-oxaloacetic transaminase
SGPT Serum glutamic pyruvic transaminase
ssDNA Single stranded DNA
TNF-α Tumor necrosis factor-α

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